

Education

- 2015-2020 **Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering**
Indian Institute of Technology Madras, Chennai, India CGPA: 8.78
- 2015 **XII - Karnataka Board, KLE Society's Independent PU College, Bangalore** 97.30 %
- 2013 **X - ICSE, B P Indian Public School, Bangalore** 96.33%

Professional Experience

- Software Engineer, Level IV, Google LLC, New York**
- Dec 2022 - Present **Software Engineer, Level IV, Google India Pvt Ltd, Bangalore**
- Aug 2020 - Nov 2022 { Developing intelligent features for the Google Workspace Editors (Docs, Slides, Keep, etc) using my expertise on the products' client-side software, supporting tools and libraries, and natural language processing infrastructure.
 { Using cutting-edge frontend tools like Web Assembly and Emscripten, and Google-internal technologies like j2Cl, client-side cross-platform frameworks and build systems, to develop user-facing features such as spellcheck in encrypted documents for five languages and writing style suggestions for English text.
 { Formulating technical designs for independent end-to-end problems, driving cross-team collaboration, upholding software reliability practices, technical-debt resolution and documentation, and proactively identifying areas of future work.
 { Guiding junior engineers on programming and software design tasks to enable timely delivery of products to customers.
- May 2019 - July 2019 **Software Engineering Intern, Google India Pvt Ltd, Bangalore**
- Worked on the Editors client-side software infrastructure to develop a user interface with control options to undo or provide feedback on the correction and a logging framework, for the Google Docs text auto-correction feature.
- May 2018 - July 2018 **Research Intern, Big Data Experience Labs, Adobe Research, Bangalore**
- Developed a mobile application for Text to Scene Conversion in Augmented Reality, based on novel research techniques for prediction of three-dimensional object sizes and positions from textual features.

Research Experience

- Sep 2019 - May 2020 **Paraphrase Generation with a Bilingual Model and Continuous Embeddings** *Master's Thesis, Language Technologies Institute, Carnegie Mellon University*
- Machinated a novel technique for paraphrase generation using the [von Mises-Fisher \(vMF\) Loss](#) on a transformer network, and showed that it produces superior paraphrases as compared to the log-likelihood model by employing bilingual data to induce zero-shot paraphrasing, guided by [Prof. Yulia Tsvetkov](#).
- May 2017 - July 2017 **Cognitive Approach to Natural Language Processing**
- Research Intern, Department of Computer Science and Automation, Indian Institute of Science (IISc), Bangalore*
- Developed a cognitive text parser that combines syntactic and semantic approaches, to process textual data into cognitive structural representations, to be used as a feature extractor for downstream NLP tasks, and demonstrated the correlation of the extracted cognitive features with semantic and syntactic text features, guided by [Prof. Veni Madhavan](#).

Publications and Patents

- [Publication and Poster] **Improving the Diversity of Unsupervised Paraphrasing with Embedding Outputs (Paper, Poster)**
Monisha Jegadeesan, Sachin Kumar, John Wieting, Yulia Tsvetkov
 In [Workshop on Multilingual Representation Learning](#),
 The 2021 Conference on Empirical Methods in Natural Language Processing ([EMNLP 2021](#))
- [Publication and Poster] **Adversarial Demotion of Gender Bias in Natural Language Generation (Paper, Poster)**
Monisha Jegadeesan
 In [ACM CODS-COMAD 2020](#) - Young Researchers' Symposium
- [Poster] **ARComposer: Authoring Augmented Reality Experiences through Text (Poster)**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
 In ACM User Interface Software and Technology Symposium 2019 ([ACM UIST 2019](#))
- [Filed Patent] **Visualizing Natural Language through 3D Scenes in Augmented Reality**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
 Filed at the US PTO (Application Number: 16/247,235)

Projects

Real-Time AI Chatbot for Customer Support (2024)

Developed and refined an AI-powered chatbot to provide instant, accurate responses to customer queries by leveraging company knowledge bases. Implemented retrieval-augmented generation (RAG) pipelines and created a vector database for efficient content tagging and organization using LangChain, OpenAI embeddings, and ChromaDB. Validated algorithms and code to ensure technical accuracy, significantly reducing human agent workload by 35% and improving response time from minutes to instant replies, demonstrating strong skills in AI integration and software validation.

Tech: Python, LangChain, OpenAI API, ChromaDB

Customer Sentiment Analysis Using NLP (2024)

Built and tested natural language processing models to analyze customer reviews for sentiment classification and insights extraction. Performed prompt refinement and code debugging to enhance model clarity and effectiveness. Cleaned and tokenized text data, engineered features with TF-IDF, and validated classification algorithms using Logistic Regression, achieving 89% accuracy. Provided structured feedback on model outputs and clearly documented technical solutions to support continuous benchmarking of model capabilities.

Tech: Python, NLTK, TF-IDF, Logistic Regression

Employee Attrition Predictive Analytics (2024)

Conducted thorough data analysis and algorithm validation to model drivers of employee attrition from complex workforce datasets. Created detailed dashboards for HR stakeholders using Power BI, facilitating data-driven insights and policy changes that reduced attrition by 12%. Refined analysis prompts and structured feedback on model performance and clarity, supporting benchmarking efforts and ensuring the highest standards of technical accuracy and documentation throughout the project lifecycle.

Tech: Python, SQL, Power BI, Logistic Regression

Website Traffic Analytics and Conversion Optimization (2024)

Analyzed website traffic patterns to identify key factors affecting user engagement and conversion rates. Developed and validated funnel analysis dashboards, tagging and organizing content by topic and traffic segments to enable precise insights. Provided structured technical feedback and refined code to improve model clarity and reporting accuracy, resulting in a 17% increase in conversion rates and a 10% reduction in bounce rates, demonstrating strong proficiency in analytics, data validation, and technical communication.

Tech: Google Analytics, SQL, Power BI

About :

Experienced Software Developer with over 2 years of expertise in software engineering and AI-driven systems. Skilled in reviewing and refining AI-generated prompts, responses, and code to ensure technical accuracy and solution clarity. Proficient in Python, JavaScript, and Java, with strong experience in debugging, testing, and validating algorithms and software concepts. Adept at providing structured feedback and organizing content by topic, difficulty, and language. Passionate about supporting benchmarking efforts to assess model capabilities and delivering high-quality, scalable AI-powered applications. Committed to technical writing with strong attention to detail and leveraging AI and machine learning technologies to build intelligent, impactful software solutions.

